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## **DH22TIN07 LESSON 1: Library management -**

system - Identify candidate classes (Books, Readers, Loans, Employees, Publishers) + Books: main entity, book information management.

+ DocGia: person who borrows books.

+ PhieuMuon: records borrowing/returning.

+ NhanVien: system manager (librarian, warehouse keeper...).

+ NhaXuatBan: book source information.

- Properties and behavior (methods) of each class + Book

o Attributes: maSach (PK, string), nameSach (string), tacGia (string), namXuatBan (int), maNXB (FK), soLuongTon (int), trangThai (string: "Available"/"On loan").

o Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(), capNhatTonKho().

+ DocGia o Attributes: maDocGia (PK, string), hoTen (string), daySinh (date), diaChi (string), soDienThoai (string), dayDangKy (date), gioiTinh (boolean).

o Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon(). + PhieuMuon o Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK), maNhanVien (FK), nhaMuon (date), ngoiHenTra (date), nhaTraThucTe (date), thanhTrang (string: "On loan"/"Paid"/"Overdue").

o Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhieuPhat(). + NhanVien o Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), date of birth (date), soDienThoai (string).

o Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().()

+ NhaXuatBan o Attributes: nameNXB (PK, string), nameNXB (string), address (string), email (string).

o Behavior: addNXB (), capNhatNXB ().

- Relationships: ☐Books and Publishers: 1 book – 1 publisher

☐☐ Readers and Loans: 1 reader – many loans ☐☐ Books and Loans: 1 book – many loans ☐☐ Employees and Loans: 1 employee – many loans **LESSON 2: Online sales management system** - Identify main use cases

Main actors:

+ Customer (Customers / Potential Customers).

+ NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

☐☐ DangKyTaiKhan (Customer registration).

☐☐ DangNhap (Sign in).

☐☐ See SanPham (Search / View product list).

☐☐ ThemVaoGioHang (Add to cart).

☐☐ DatHang (Order).

☐☐ ThanhToan (Online payment / COD).

☐☐ View DonHang (View order history).

Admin / NguoiQuanLy:

☐☐ QuanLySanPham (Add/Edit/Delete products).

☐☐ QuanLyDonHang (View/Confirm/Cancel order).

☐☐ QuanLyKhachHang (Customer information management).

☐☐ QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

☐☐ ThanhToan includes XacNhanDonHang.

☐☐ DatHang extends ThanhToan (if paying online).

- From use case, find necessary classes ☐☐ Customer (Customer): from register, view orders.

☐☐ SanPham (Product): product management.

☐☐ GioHang (Cart): temporarily store products before ordering.

☐☐ Don Hang (Order): from order, payment.

☐☐ ChiTietDonHang (OrderDetail): product details in the order.

☐☐ ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).

☐☐ Admin or Personnel (inherited from User if there is a user system).

- Analyze the relationship between 2 classes (specify inheritance if 2 any) □  
Customer and Customer: 1 customer – many 2 orders (abstract class) ← inherits 2  
2 ← KhachHang and Admin (KhachHang and 2 Admin are both the same type of User, sharing the same account, password, full name).

### Lesson 3: Convert class diagrams into data tables

#### 1. MonHoc table

| No | School name | Data Types   | Bind |    |          |        |
|----|-------------|--------------|------|----|----------|--------|
|    |             |              | PK   | FK | Not NULL | Unique |
| 1  | maMon       | Nvarchar(10) | x    |    | x        | x      |
| 2  | nameMon     | Nvarchar(50) |      |    | x        |        |
| 3  | soTinChiLT  | Int          |      |    | x        |        |
| 4  | soTinChiTH  | Int          |      |    | x        |        |

#### 2. Science Table

| No | School name | Data Types   | Bind |    |          |        |
|----|-------------|--------------|------|----|----------|--------|
|    |             |              | PK   | FK | Not NULL | Unique |
| 1  | maKhoaHoc   | Nvarchar(10) | x    |    |          | x      |
| 2  | tenKhoaHoc  | Nvarchar(50) |      |    | x        |        |
| 3  | ngayTao     | Date         |      |    | x        |        |